10.1 Measuring an angle in radians

Definition and results

The unit circle is the circle with centre O and radius 1, oriented anticlockwise. A radian corresponds to an arc of length 1 on this circle. One turn is 2 π radians, half a turn π. On a circle of radius r, an arc length is $r\theta$ for a positive angle θ expressed in radians.

Status and justification. Definitions and proportionality between angle at the centre and arc length. The conversion is 180 $^\circ = \pi$ radians; multiply the degrees by $\frac{\pi}{180}$.

Worked example. $60^\circ = \frac{\pi}{3}$; $\frac{3\pi}{2} = 270^\circ$. An angle of $\frac{\pi}{3}$ on a circle of radius 6 cm gives an arc of 2 π cm.

Watch out

The length of an arc is not that of the chord. In trigonometric calculations, check whether the tool is in degree or radian mode.

Immediate application. Convert 45 $^\circ$, 120° ; give the arc with radius 2 m and angle $\frac{3\pi}{4}$.

10.2 Wrapping and the image of a real number

Definition and results

To each real number x corresponds a point M on the circle obtained by a directed path of length x from I(1;0). Two real numbers have the same image if their difference is an integer multiple of 2 π. A negative number indicates travel in the clockwise direction.

Status and justification. Adding or removing a whole number of turns keeps the end point. Conversely, returning to the same point requires a whole number of turns.

Worked example. $\frac{13\pi}{3} = 4\pi + \frac{\pi}{3}$ has the same image as $\frac{\pi}{3}$. $-\frac{\pi}{2}$ has the same image as $\frac{3\pi}{2}$: point $(0; -1)$.

Watch out

Add π gives the opposite point, not the same point.

Immediate application. Plot the images of $\frac{7\pi}{2}$ and $-\frac{5\pi}{3}$ by reducing them to one turn.

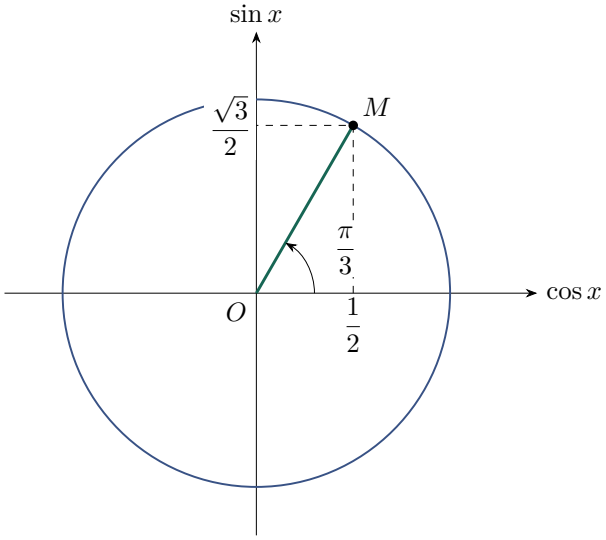
10.3 Sine, cosine and standard values

Definition and results

If M is the image of x, $M = (\cos x; \sin x)$. Hence $-1 \leq \cos x, \sin x \leq 1$ and $\cos^2 x + \sin^2 x = 1$.
For $x = 0, \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{\pi}{2}$, the cosines are 1 $, \frac{\sqrt{3}}{2}, \frac{\sqrt{2}}{2}, \frac{1}{2}$, 0 and the sines 0 $, \frac{1}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, 1$.

Status and justification. À $\frac{\pi}{4}$, the two coordinates are equal and positive; Pythagoras gives $2a^2 = 1$, therefore $a = \frac{\sqrt{2}}{2}$. An equilateral triangle with side 1, cut by its altitude, has half-base $\frac{1}{2}$ and height $\frac{\sqrt{3}}{2}$: $\cos(\frac{\pi}{3}) = \frac{1}{2}$, $\sin(\frac{\pi}{3}) = \frac{\sqrt{3}}{2}$. The complementary angle gives $\frac{\pi}{6}$.

Worked example. The point at angle $\frac{\pi}{3}$ is $(\frac{1}{2}; \frac{\sqrt{3}}{2})$. The scalar product of two unit vectors making this angle equals its cosine, so $\frac{1}{2}$. This links the geometry chapters to the circle.



For the angle $\frac{\pi}{3}$, the projections of M give its cosine and sine directly. The fractions are placed outside the lines of the drawing.

Watch out

The cosine is the x-coordinate, the sine the y-coordinate. The identity gives an absolute value: the quadrant determines the sign.

Immediate application. If $\cos x = \frac{3}{5}$ and the image of x is in the fourth quadrant, give $\sin x$.

10.4 Related angles and symmetries

Definition and results

The symmetries of the circle give $\cos(-x) = \cos x$, $\sin(-x) = -\sin x$; $\cos(\pi - x) = -\cos x$, $\sin(\pi - x) = \sin x$; $\cos(\pi + x) = -\cos x$, $\sin(\pi + x) = -\sin x$. Adding 2 $k\pi$, k integer, does not change neither sine nor cosine.

Status and justification. Each equality can be read from the coordinates of the symmetric points: the horizontal axis for $-x$, vertical axis for $\pi - x$, centre O for $\pi + x$. This is not a list of formulas without justification.

Worked example. $\cos(\frac{5\pi}{6}) = -\frac{\sqrt{3}}{2}$, $\sin(\frac{5\pi}{6}) = \frac{1}{2}$ by symmetry of $\frac{\pi}{6}$. $\cos(-\frac{\pi}{4}) = \frac{\sqrt{2}}{2}$, $\sin(-\frac{\pi}{4}) = -\frac{\sqrt{2}}{2}$.

Watch out

Two angles with the same cosine do not necessarily have the same sine. Reduce the angle, then read the signs.

Immediate application. Calculate the two coordinates of the images of $\frac{4\pi}{3}$ and $-\frac{\pi}{4}$.

10.5 Approximating pi and reusing ideas

Definition and results

A regular n-sided polygon inscribed in the unit circle has semi-perimeter $n\sin(\frac{\pi}{n})$; a circumscribed polygon has semi-perimeter $n\tan(\frac{\pi}{n})$, with $\tan\theta = \frac{\sin\theta}{\cos}\theta$ here. Archimedes' method doubles the number of sides and calculates the new lengths by Pythagoras, without knowing pi in advance.

Status and justification. The perimeter of the inscribed polygon is less than that of the circle, and that of the circumscribed one is greater: geometric bounds assumed. If c is the chord of one side, set $h = \sqrt{1 - \frac{c^2}{4}}$; the chord of the half-angle is $\frac{c}{\sqrt{2(1+h)}}$. This formula comes from Pythagoras and avoids subtracting two nearly equal numbers.

Worked example. For the inscribed hexagon, c=1 and semi-perimeter 3. The circumscribed one gives 2 $\sqrt{3} \approx$ 3,4641. Doubling the number of sides tightens the bounds. The given program starts from n=6 and c=1.

Watch out

Using a sine function directly with pi already known would check a calculation, but would not construct an independent approximation of pi.

Immediate application. For n=6 and c=1, calculate the new chord after doubling.